

AI Final Task

Table 1 — Direct Comparison

#	Question	Without Hybrid Search	Hybrid Search
1	How many slots does the 1830 PSS-8 shelf provide, and what is its rack-unit (RU) footprint?	8-slot SWDM platform, 3-RU	8-slot SWDM platform, 3-RU
2	What rack-unit footprint does the 1830 PSS-32 shelf have, and how many slots does it provide?	14-RU, 32 slots	14-RU, 32-slot high-capacity core SWDM platform
3	What are the two software load-lines supported by the 1830 PSS system?	SWDM software, OCS software	SWDM software, OCS software
4	Which fan unit(s) are used on the 1830 PSS-16II shelf?	16FAN2, 16FAN2C	16FAN2, 16FAN2C
5	Which fan units are supported on the 1830 PSS-32 shelf?	FAN, FAN32H	FAN, FAN32H
6	Which power filter cards are supported on the 1830 PSS-8 shelf?	No information found	8DC30, 8DC30T, 8DC30T2, 8AC7
7	What is the required horizontal rack aperture for mounting a 1830 PSS-8 shelf, and which common aperture size is explicitly NOT supported?	450.85 mm; 444.5 mm not supported	450.85 mm; 444.5 mm not supported
8	What is the maximum optical reach, in kilometers, of the 1830 PSS-8 shelf without amplification	No information found	No information found

Table 2 — Answer + Retrieved Chunk Content + Success

#	Method	Question	Answer / Retrieved Evidence	Chunk (source)	Success?
1	Without Hybrid search	How many slots does the 1830 PSS-8 shelf provide, and what is its rack-unit (RU) footprint?	The 1830 PSS-8 shelf provides an 8-slot SWDM platform in 3-RU footprint, with additional distributed switching functionality. 1830 PSS-8 can support up to four full-height or up to eight half-height universal I/O cards when that slot is also equipped with a half-slot adapter.	Chunk 5 — Page 35 / PSS-8 introduction (retrieved output also showed other distractor chunks)	Success
1	Hybrid search	How many slots does the 1830 PSS-8 shelf provide, and what is its rack-unit (RU) footprint?	The 1830 PSS-8 shelf provides an 8-slot SWDM platform in 3-RU footprint, with additional distributed switching functionality. 1830 PSS-8 can support up to four full-height or up to eight half-height universal I/O cards when that slot is also equipped with a half-slot adapter.	Chunk 5 — Page 35 / PSS-8 introduction (retrieved output)	Success
2	Without Hybrid search	What rack-unit footprint does the 1830 PSS-32 shelf have, and how many slots does it provide?	The 1830 PSS-32 shelf provides a 32-slot high-capacity and high-scalability core SWDM platform in 14-RU footprint. The 1830 PSS-32 can support up to thirty two full-height or up to sixteen half-height universal I/O cards when that slot is also equipped with a half-slot adapter (HSLAD). In addition slots for two power filters, two controller cards are supported.	Chunk 5 — Page 50 / 2.5.1 Introduction	Success
2	Hybrid search	What rack-unit footprint does the 1830 PSS-32 shelf have, and how many slots does it provide?	The 1830 PSS-32 shelf provides a 32-slot high-capacity and high-scalability core SWDM platform in 14-RU footprint. The 1830 PSS-32 can support up to thirty two full-height or up to sixteen half-height universal I/O cards when that slot is also equipped with a half-slot adapter (HSLAD). In	Chunk 5 — Page 50 / 2.5.1 Introduction	Success

			addition slots for two power filters, two controller cards are supported.		
3	Without Hybrid search	What are the two software load-lines supported by the 1830 PSS system?	The system supports two software load-lines: SWDM software and OCS software. Each software load-line may support only a subset of the shelves. Shelf names indicate the load-line as PSS-xy (SWDM) or PSS-xy (OCS).	Chunk 1 — Page 1 / System concept overview	Success
3	Hybrid search	What are the two software load-lines supported by the 1830 PSS system?	The system supports two software load-lines: SWDM software and OCS software. Each software load-line may support only a subset of the shelves. Shelf names indicate the load-line as PSS-xy (SWDM) or PSS-xy (OCS).	Chunk 1 — Page 1 / System concept overview	Success
4	Without Hybrid search	Which fan unit(s) are used on the 1830 PSS-16II shelf?	The 1830 PSS-16II FAN has 10 FAN units. Each FAN unit is powered separately. The 1830 PSS-16II FAN can support a M+1 FAN unit protection scheme.	Chunk 2 — Page 104	Success
4	Hybrid search	Which fan unit(s) are used on the 1830 PSS-16II shelf?	Retrieved context included the PSS-16 FAN section and the PSS-16II FAN details: the PSS-16II FAN has 10 FAN units and supports M+1 protection. The final answer was 16FAN2 and 16FAN2C.	Chunk 2 — Page 104 (plus retrieved PSS-16 fan chunks)	Success
5	Without Hybrid search	Which fan units are supported on the 1830 PSS-32 shelf?	The retrieved evaluation output returned the answer: FAN and FAN32H. The relevant PSS-32 fan section identifies the Fan Unit (FAN or FAN32H) as mounted on the PSS-32 shelf.	PSS-32 fan section — Page 153 / Fan Unit (FAN or FAN32H)	Success
5	Hybrid search	Which fan units are supported on the 1830 PSS-32 shelf?	The retrieved evaluation output returned the answer: FAN and FAN32H. The relevant PSS-32 fan section identifies the Fan Unit (FAN or FAN32H) as mounted on the PSS-32 shelf.	PSS-32 fan section — Page 153 / Fan Unit (FAN or FAN32H)	Success
6	Without Hybrid search	Which power filter cards are supported on the 1830 PSS-8 shelf?	Retrieved chunks were mainly around the PSS-8 power-filter section but the generated answer incorrectly said that the documents did not contain the information. The document actually contains the PSS-8 power-filter card names.	Chunks around Pages 112–113 (retrieval did not use the most useful chunk for the final answer)	Fail
6	Hybrid search	Which power filter cards are supported on the 1830 PSS-8 shelf?	The PSS-8 power-filter section is explicitly titled “Power filter cards (8DC30, 8DC30T, 8DC30T2, 8AC7)” and states that the PSS-8 shelf supports both AC and DC Power filter cards.	Chunk — Page 110 / 2.20 PSS-8 Power filter cards; supporting details also appear on Pages 112–113	Success
7	Without Hybrid search	What is the required horizontal rack aperture for mounting a 1830 PSS-8 shelf, and which common aperture size is explicitly NOT supported?	The following rack types are supported: 19-inch EIA rack, ETSI rack, and 23-inch ANSI rack. The horizontal aperture must be 450.85 mm (17.75 in). Racks with 444.5 mm (17.5 in) horizontal aperture are not supported.	Chunk 1 — Page 36 / 2.2.2 Rack mounting options	Success
7	Hybrid search	What is the required horizontal rack aperture for mounting a 1830 PSS-8 shelf, and which common aperture size is explicitly NOT supported?	The following rack types are supported: 19-inch EIA rack, ETSI rack, and 23-inch ANSI rack. The horizontal aperture must be 450.85 mm (17.75 in). Racks with 444.5 mm (17.5 in) horizontal aperture are not supported.	Chunk 1 — Page 36 / 2.2.2 Rack mounting options	Success
8	Without Hybrid search	What is the maximum optical reach, in kilometers, of the 1830 PSS-8 shelf without amplification?	Retrieved chunks included unrelated notes, PSS-16II information, rack information, and system overview, but no value for maximum optical reach. The final answer stated that the provided documents did not contain the requested reach information.	Chunks 1–5— especially Page 10, Page 47, Page 36, Page 34, Page 1, Page 15, Page 3	Success*
8	Hybrid search	What is the maximum optical reach, in kilometers, of the 1830 PSS-8 shelf without amplification?	Hybrid retrieval returned the same general set of unrelated/context chunks and did not retrieve a chunk containing a maximum-reach value. The final answer stated that the provided documents did not contain the requested reach information.	Chunks 1–5 — especially Page 10, Page 47, Page 36, Page 34, Page 1, Page 15, Page 3	Success*